WEEK 2 CYBERSECURITY PROJECT REPORT

Footprinting, Reconnaissance and Network Scanning

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Operating Environment: Kali Linux running in VirtualBox
Target Domain: networkwalks.com
Local Lab Network: 10.0.0.0/24

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1. Introduction

This report documents the practical cybersecurity activities completed during Week 2 of the Networkwalks Cybersecurity Internship.

The exercises focused on footprinting, reconnaissance and network scanning using tools available in Kali Linux and the Zenmap graphical interface.

Three project modules were completed:

- W2-PM1: Footprinting and reconnaissance using multiple Kali tools.
- W2-PM4: Footprinting using theHarvester.
- W2-PM5: Network scanning using Zenmap.

The exercises involved collecting publicly available information about the authorized target domain and identifying live hosts within the local virtual network. The results were documented through terminal outputs, screenshots and a network topology PDF.

2. Objectives

The objectives of the practical exercises were to:

1. Retrieve domain registration information using WHOIS.
2. Identify web technologies and server information using WhatWeb.
3. Resolve domain names to IP addresses using DNS lookup tools.

4. Examine HTTP response headers using cURL.
5. Identify the presence of a Web Application Firewall using WAFW00F.
6. Enumerate DNS records using DNSRecon.
7. Discover publicly indexed emails, hosts and subdomains using theHarvester.
8. Identify live hosts, IP addresses and MAC addresses within the authorized local subnet.
9. Generate and save a graphical network topology using Zenmap.

3. Authorization and Scope

The exercises were conducted under the authorization and scope provided in the Networkwalks Week 2 project materials.

The authorized targets were:

- The Networkwalks public website and its associated DNS records.
- The student's own local area network within the controlled lab environment.

The permitted activities included footprinting, reconnaissance, network scanning and host discovery.

Exploitation, unauthorized access, denial-of-service activities, brute-force attacks and testing outside the approved scope were excluded.

Scope adjustment: The PM4 manual referenced microsoft.com as its example target. However, Microsoft was not included in the authorization scope used for this exercise. Therefore, networkwalks.com was used for both theHarvester tasks to maintain compliance with the authorization boundaries.

4. Testing Environment and Tools

4.1 Environment

The practical exercises were performed using Kali Linux in Oracle VirtualBox.

The local network configuration was:

Configuration	Value
Network interface	eth0
Local IP address	10.0.0.2
Subnet	10.0.0.0/24

Default gateway	10.0.0.1
Network address	10.0.0.0
Usable host range	10.0.0.1–10.0.0.254

4.2 Tools Used

Tool	Purpose
WHOIS	Retrieve domain registration information
WhatWeb	Identify website technologies
Nslookup	Resolve domain names to IP addresses
cURL	Retrieve HTTP response headers
WAFW00F	Identify the presence of a Web Application Firewall
DNSRecon	Enumerate DNS records
theHarvester	Collect publicly available emails, hosts and domain information
Nmap	Discover live hosts within the local subnet
Zenmap	Graphical interface for Nmap and network topology visualization

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5. W2-PM1: Footprinting and Reconnaissance Using Multiple Kali Tools

5.1 WHOIS – Domain Registration Information

Command executed:

```
whois networkwalks.com
```

Results:

Information	Observed result
Domain	NETWORKWALKS.COM

Registrar	GoDaddy.com, LLC
Creation date	6 November 2019
Last updated	12 November 2025
Expiry date	6 November 2027
Name server 1	NS6135.HOSTGATOR.COM
Name server 2	NS6136.HOSTGATOR.COM
DNSSEC	Unsigned

Analysis:

The WHOIS lookup identified GoDaddy.com, LLC as the domain registrar. The domain was registered on 6 November 2019 and is scheduled to expire on 6 November 2027.

The listed name servers belong to the HostGator naming infrastructure. This provides information about the domain's DNS delegation.

The domain's DNSSEC status was reported as unsigned.

5.2 WhatWeb – Website Technology Fingerprinting

Command executed:

```
whatweb networkwalks.com
```

Results:

The tool identified the following technologies and information:

- Web server: Apache.
- IP address: 192.232.216.135.
- Website title: Networkwalks Academy.
- Content management system: WordPress 7.1.4 (as reported by WhatWeb).
- WordPress Download Manager: 3.3.76.
- jQuery: 3.7.1.
- Bootstrap: 7.1.4 (as reported by WhatWeb).
- Google Tag Manager.
- HTML5 and Open Graph protocol.
- Contact email: info@networkwalks.com.

- Cookie detected: `__wpdm_client`.

The HTTP version of the website returned a 301 Moved Permanently response and redirected to HTTPS. The HTTPS version returned 200 OK.

Analysis:

WhatWeb successfully identified the website's web server, content management system and several client-side technologies.

The detected IP address was 192.232.216.135. The results also showed that the website redirects HTTP requests to HTTPS.

These findings provide a basic technology profile of the target website. The reported software versions are fingerprinting results and were not independently verified.

5.3 Nslookup – DNS Resolution

Command executed:

```
nslookup networkwalks.com
```

Results:

- DNS resolver: 10.0.0.1.
- Domain: networkwalks.com.
- Resolved IP address: 192.232.216.135.
- Response type: Non-authoritative answer.

Analysis:

The lookup successfully resolved networkwalks.com to 192.232.216.135.

The result matched the IP address identified by WhatWeb, providing consistency between the two tools.

The response was identified as non-authoritative, meaning the answer was supplied by the DNS resolver rather than directly by an authoritative nameserver.

5.4 cURL – HTTP Response Headers

Command executed:

```
curl -I https://networkwalks.com
```

Results:

The server returned:

- HTTP status: HTTP/2 200.
- Server: Apache.
- Content type: text/html; charset=UTF-8.
- Response date: 11 November 2026.
- Cache-related headers: `x-endurance-cache-level` and `x-nginx-cache`.
- Referrer policy: `no-referrer-when-downgrade`.
- Cookie: `__wpdm_client`, with Secure and HttpOnly attributes.
- WordPress REST API links were present in the response headers.

Analysis:

The cURL command successfully retrieved the HTTP response headers without downloading the full webpage.

The HTTP/2 200 response indicated that the HTTPS request was successful.

The headers also exposed server information, caching details and references to WordPress REST API endpoints. These details contribute to understanding the website's publicly visible technology configuration.

Cookie values were excluded from this report to avoid unnecessarily exposing session-related information.

5.5 WAFW00F – Web Application Firewall Detection

Command executed:

```
wafw00f https://networkwalks.com
```

Results:

- WAFW00F version: 2.4.2.
- Detected WAF: ModSecurity (SpiderLabs).
- Number of requests: 2.

Analysis:

WAFW00F identified ModSecurity (SpiderLabs) as the Web Application Firewall protecting the target website.

This indicates that a WAF fingerprint was detected during the test. The exercise did not assess the effectiveness of its rules or attempt to bypass the protection.

5.6 DNSRecon – DNS Enumeration

Command executed:

```
dnsrecon -d networkwalks.com
```

Results:

The tool identified the following DNS information:

Record or information	Observed result
SOA	ns6135.hostgator.com
NS 1	ns6135.hostgator.com – 50.87.144.87
NS 2	ns6136.hostgator.com – 192.232.216.131
BIND version	9.16.23-RH, reported for both nameserver IPs
MX	mail.networkwalks.com – 192.232.216.135
A record	networkwalks.com – 192.232.216.135
TXT	SPF record detected
TXT	Google site verification record detected
SRV	Eight autodiscover service records detected

The SPF record returned was:

```
v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
```

The tool also reported:

```
No answer for DNSSEC query for networkwalks.com
```

Analysis:

DNSRecon identified the domain's name servers, mail server, A record, TXT records and service records.

The SPF record provided information about the domain's stated email-sending policy. The SRV records revealed autodiscover-related service configuration associated with cPanel email discovery.

The DNSSEC query returned no answer, which was consistent with the unsigned DNSSEC status reported by WHOIS. The query error alone does not establish the precise reason for the missing response.

PM1 Summary:

The six tools provided complementary information about domain registration, DNS infrastructure, website technologies, HTTP headers and WAF presence.

The main IP address identified across the WHOIS-related reconnaissance results, WhatWeb, Nslookup and DNSRecon was 192.232.216.135.

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6. W2-PM4: Footprinting Using theHarvester

TheHarvester was used to collect publicly available information associated with networkwalks.com.

Two searches were conducted using different source configurations.

6.1 Task 1 – Baidu Search

Command executed:

```
theHarvester -d networkwalks.com -l 1000 -b baidu
```

Results:

- Tool version: 4.11.1.
- Search source: Baidu.
- IP addresses found: None.
- Email addresses found: None.
- People found: None.
- Hosts found: None.

A Python SyntaxWarning concerning a return statement inside a finally block appeared during execution. The tool continued to the search operation and returned the results above.

Analysis:

The Baidu search did not return any email addresses, IP addresses, people or hosts associated with the target domain.

This represents a zero-result search from that particular source. It does not establish that the domain has no publicly discoverable information.

6.2 Task 2 – Search Using All Available Sources

Command executed:

```
theHarvester -d networkwalks.com -l 50 -b all
```

Results:

Result category	Tool output
IP addresses	3 reported by the summary
Hosts	23 reported
Subdomains	3 found through DNS fallback
Email addresses	None found
People	None found
LinkedIn users	None found
LinkedIn links	0
ASNs	1
Interesting URLs	1

The explicitly displayed IP address was:

```
192.232.216.135
```

The ASN identified was:

```
AS31898
```

The interesting URL returned was:

```
https://networkwalks.com/
```

Examples of host entries included:

- autodiscover.networkwalks.com
- cpanel.networkwalks.com
- cpcalendars.networkwalks.com
- cpcontacts.networkwalks.com
- ftp.networkwalks.com

- mail.networkwalks.com
- webdisk.networkwalks.com
- webmail.networkwalks.com

The output also contained aliases and IPv6-mapped representations. Therefore, the 23 host entries should not be interpreted as 23 unique subdomains.

API and source limitations:

Several search sources returned missing or invalid API-key messages. These included services such as Shodan, Censys, Hunter, BuiltWith, VirusTotal and others.

Some sources operated without API keys or through fallback methods. The tool reported three subdomains discovered using DNS fallback.

The Hudson Rock section displayed a domain statistic of 101 total compromised accounts/users. However, its own processing result reported zero hosts, zero IPs, zero emails and zero stealer results. This statistic was not independently verified and is not treated as confirmation of a breach involving the target organization.

Analysis:

The all-sources search returned more information than the Baidu-only search. It identified host entries, DNS-related information, an ASN and an interesting URL.

However, missing API credentials limited the coverage of several sources. The results should therefore be treated as partial reconnaissance findings rather than a complete inventory of the target's public information.

PM4 Summary:

TheHarvester demonstrated how different public sources can contribute to domain footprinting. The Baidu-only search returned no findings, while the all-sources search produced host and DNS-related results despite API limitations.

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7. W2-PM5: Zenmap-Based Network Scanning

7.1 Environment Identification

The following commands were used to identify the local network configuration:

```
ip -br addr
```

```
ip route
```

The interface output showed:

- Interface: eth0.
- IPv4 address: 10.0.0.2/24.

The routing table showed:

- Default gateway: 10.0.0.1.
- Connected subnet: 10.0.0.0/24.

Nmap version 7.95 was available in the Kali environment.

7.2 Host Discovery

Command executed:

```
sudo nmap -sn 10.0.0.0/24
```

The `-sn` option was used for host discovery without performing a port scan.

Results:

Nmap reported 256 IP addresses scanned and three hosts up.

Host	IP address	Status
1	10.0.0.1	Up
2	10.0.0.2	Up
3	10.0.0.3	Up

Total live hosts: 3

7.3 MAC Address Identification

The Nmap scan displayed MAC addresses for two discovered hosts:

IP address	MAC address
10.0.0.1	52:54:00:12:35:00

10.0.0.3 52:54:00:12:35:00

The Kali machine's MAC address was obtained separately using:

```
ip link show eth0
```

The returned MAC address was:

```
08:00:27:55:c4:28
```

The completed results were:

IP address	Device identification	MAC address
10.0.0.1	Gateway	52:54:00:12:35:00
10.0.0.2	Kali Linux machine	08:00:27:55:c4:28
10.0.0.3	Discovered host	52:54:00:12:35:00

Observation:

The scan returned the same MAC address for 10.0.0.1 and 10.0.0.3. This was recorded as observed. The result may be related to the virtual networking environment or address-resolution behaviour and was not independently investigated further.

7.4 Network Topology

Zenmap was used to display the discovered network topology and export it in PDF format.

The exported topology showed the discovered network addresses:

- 10.0.0.1
- 10.0.0.2
- 10.0.0.3

It also displayed localhost, which represents the local machine's loopback identity rather than an additional live LAN host.

The exported topology PDF was retained as supporting evidence.

PM5 Summary:

The host-discovery scan identified three live hosts within the 10.0.0.0/24 virtual network. Their IP addresses and available MAC-address information were documented, and the network topology was exported using Zenmap.

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8. Overall Observations

The three completed modules provided different but complementary views of the target domain and the local virtual network.

1. WHOIS and DNSRecon revealed domain registration, nameserver, mail and DNS configuration information.
2. WhatWeb and cURL identified the web server, WordPress-related technologies and HTTP response details.
3. WAFW00F identified ModSecurity (SpiderLabs) as the detected WAF.
4. TheHarvester returned host and DNS-related information, although several sources were unavailable because of missing API credentials.
5. Nmap and Zenmap identified three live hosts in the local virtual subnet and provided a graphical representation of the network.

The findings demonstrate how reconnaissance and host discovery can help build an initial understanding of an environment before any deeper security assessment is considered.

No exploitation or vulnerability validation was performed during these exercises.

9. Recommendations

Based on the observations, the following defensive recommendations are suggested:

1. **Review exposed technology information:** Maintain supported versions of WordPress, WordPress Download Manager, Apache and associated libraries. Independently verify installed versions before making upgrade decisions.
2. **Review DNS configuration:** Periodically audit public DNS records, including A, MX, TXT, NS and SRV records, to ensure that only necessary information and services are exposed.
3. **Review DNSSEC:** Evaluate whether DNSSEC signing should be enabled and properly maintained.
4. **Review email security:** Assess the SPF policy and consider appropriate DKIM and DMARC configuration.
5. **Review HTTP security configuration:** Evaluate response headers and cookie settings against the organization's security requirements.
6. **Maintain WAF protection:** Keep ModSecurity rules and associated security configurations updated and monitored.

7. **Review exposed subdomains:** Confirm that discovered hostnames correspond to required services and that unnecessary services are not publicly exposed.
8. **Maintain network visibility:** Regularly inventory authorized network devices and review unexpected changes in host discovery results.

These recommendations are based on reconnaissance observations and do not imply that a vulnerability was confirmed.

10. Limitations

The exercises had the following limitations:

- Some theHarvester sources required API keys that were unavailable or invalid.
- The Baidu-only search returned no results.
- TheHarvester's reported host and IP totals did not necessarily represent unique hostnames or unique IP addresses.
- Software versions identified through WhatWeb were not independently verified.
- The Zenmap scan covered the virtual subnet visible to Kali, not necessarily the physical devices on the host computer's Wi-Fi network.
- The identical MAC addresses returned for two discovered IP addresses were not independently investigated.
- No exploitation, vulnerability validation, password attacks or denial-of-service testing was performed.

11. Conclusion

The Week 2 practical exercises successfully demonstrated the use of reconnaissance and network-scanning tools in an authorized cybersecurity learning environment.

W2-PM1 provided information about domain registration, website technologies, DNS records, HTTP headers and WAF detection. W2-PM4 demonstrated theHarvester's ability to gather publicly available domain-related information, while also showing the limitations introduced by unavailable API credentials. W2-PM5 demonstrated local network host discovery, IP and MAC address identification, and network topology visualization using Zenmap.

Overall, the exercises improved practical understanding of footprinting, DNS enumeration, web technology identification and network discovery. The collected evidence provides a documented record of the activities performed and their observed results.

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12. Appendices and Supporting Evidence

The following evidence should accompany the report:

Appendix A – W2-PM1

- A1: WHOIS screenshot and complete text output.
- A2: WhatWeb screenshot and complete text output.
- A3: Nslookup screenshot and complete text output.
- A4: cURL screenshot and complete text output.
- A5: WAFW00F screenshot and complete text output.
- A6: DNSRecon screenshot and complete text output.

Appendix B – W2-PM4

- B1: theHarvester Baidu search screenshot and complete text output.
- B2: theHarvester all-sources search screenshot and complete text output.

Appendix C – W2-PM5

- C1: Kali IP address and subnet evidence.
- C2: Nmap host-discovery screenshot and output.
- C3: Kali MAC address evidence.
- C4: Zenmap network topology PDF.

End of Report